# Turf Ground Booking System

Requirement Analysis and User Flow

Product Management Intern Assignment

Prepared by

**Md. Rahat Pervez**

rahatparvez.official@gmail.com

Prepared for RoBenDevs

September 2026

## Project Summary

This document sets out the requirements and the customer flow for a Turf Ground Booking System. The turf currently takes bookings by phone, Facebook and WhatsApp, so customers cannot see which slots are free and the team has to keep every booking by hand. The proposed system is a mobile-friendly booking page with live slot availability, together with a small admin page for the owner and staff.

- **Business problem:** customers cannot see live slot availability, and the turf has no single reliable list of bookings.
- **Proposed solution:** a mobile-friendly booking page that shows live availability and updates slot status automatically. Staff only handle exceptions such as maintenance or phone bookings.
- **First release (MVP):** view slots, book, cancel, and a simple admin page. There is no online payment yet.
- **Expected outcome:** no double bookings, fewer booking calls and messages, and one clear booking list for the team.

## What I understand about the problem

A customer who wants to play has to call or message the turf to find out whether a slot is free, and then wait for a reply. The turf team receives bookings on three channels and writes them down by hand, so a message can be missed and the same slot can be promised twice. The owner has no clear record of bookings, cancellations or how busy the turf is.

The problem in one line: customers cannot see live availability, and the turf has no single reliable list of bookings.

## Main users

- **Customer:** a player, or one person booking for a group. Wants to see free slots quickly, book without messaging, get a confirmation and cancel if plans change.
- **Turf owner or manager:** runs the turf. Wants to see all bookings in one place, block slots for maintenance and avoid double bookings.
- **Turf staff:** front-desk staff who still get phone and walk-in bookings. They need to add those bookings quickly so the system stays correct.

## What a customer needs to provide

- **Required:** full name, mobile number, date and time slot.
- **Optional:** number of players, email and a note for the turf.
- **Created by the system:** booking ID and booking status.

There is no account or password. Name and mobile number are enough, because the turf already contacts its customers by mobile.

## What the turf owner needs to manage

The owner sets these once, and the system uses them to work out availability.

- Turf details: name, address, photos, sport, facilities and rules.
- Opening hours, slot length and prices (one price, or different prices for peak hours and weekends).
- Closed days, holidays and maintenance blocks.
- The cancellation policy that customers see before they book.

The owner can also view the booking list (date, slot, customer name and mobile number, and whether the booking is confirmed or cancelled) and handle exceptions: add a phone or walk-in booking, cancel a booking when a customer asks, and block or unblock a slot.

## How availability works

The system works out slot status. Staff do not update it by hand for online bookings. A slot is available, held, booked or blocked.

- When a customer picks a slot, it is held for 5 minutes. It becomes booked when the customer confirms. It becomes available again if the customer leaves or the 5 minutes pass.
- When a booking is cancelled, the slot becomes available again on its own.
- Staff step in only for things a system cannot know: blocking a slot for maintenance or a holiday, entering a phone or walk-in booking, and cancelling for a customer who calls. A regular closure, such as every Monday morning, can be set once as a rule.

Figure 1 shows how the parts of the system fit together. Every booking goes through the booking service, which checks the database in one step, so two people cannot get the same slot.

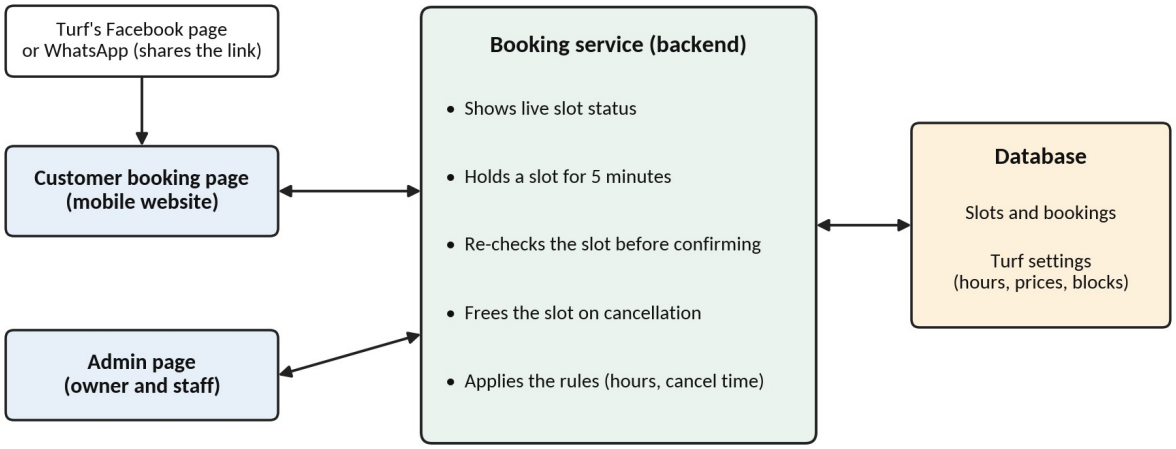



*Figure 1. System overview*

## Questions before development starts

These are the questions I would ask the owner first.

1. How many grounds does the turf have, and which sports are played?
2. What are the opening hours and the slot length? How far ahead can customers book?

3. Is the price the same for every slot, or different for peak hours and weekends?
4. Should customers pay online (advance or full) or at the turf? If online, how are refunds handled?
5. What is the cancellation policy, and is there any charge?
6. Will phone and WhatsApp bookings continue? If yes, who enters them into the system so slots stay correct?
7. If rain or maintenance forces a cancellation, does the customer get a new slot, a refund or both?
8. Who will use the admin page: only the owner, or staff as well?
9. Should this be a website or an app, and in Bangla, English or both?
10. How should customers be notified (on screen, SMS or WhatsApp), and what are the budget and launch date?

## Assumptions

Until those questions are answered, I am assuming the following.

1. One ground and one sport at launch, with fixed one-hour slots inside opening hours set by the owner.
2. Customers can book up to 30 days ahead, and one booking is one slot.
3. A booking is confirmed at once if the slot is free. The owner does not approve each booking.
4. Slot status updates automatically. The brief sets no limit that forces manual handling, so this is a normal web application with a backend and a database.
5. There is no online payment in the first release. Customers pay at the turf.
6. There are no customer accounts. Customers manage a booking with the booking ID and mobile number.
7. A customer can cancel free of charge up to 12 hours before the slot starts. The owner needs to confirm this.
8. A selected slot is held for 5 minutes.
9. It is a mobile-friendly website, in English first and Bangla later. All times are Bangladesh time.
10. One admin role in the first release. Phone and walk-in bookings are entered by the admin.

## Priorities

- **Must have (first release):** view turf details, live slot availability by date, booking with name and mobile number, automatic slot status with no double bookings, booking confirmation with a booking ID, and customer cancellation. For the admin: view bookings, block slots, add a booking and cancel a booking.
- **Should have (next):** SMS or WhatsApp confirmation and reminders, online advance payment (for example bKash or Nagad), regular closures set as a rule, and simple reports for the owner.
- **Later:** more than one ground, weekly repeat bookings, customer accounts and booking history, and offers.

## Key decisions and trade-offs

- **Availability is automatic.** Manual upkeep is the core problem, so the system updates status on booking, cancellation and hold expiry. Trade-off: more engineering work at the start, and holds and cancellations need careful handling.
- **Instant confirmation, no owner approval.** Approval would bring the manual work back. Trade-off: fake or no-show bookings are possible. The mobile number and the cancellation rule reduce this, and advance payment comes next.
- **No online payment in the first release.** It keeps the first version simple and quick to launch. Trade-off: with no deposit, no-shows are possible.
- **No customer accounts.** Fewer steps means more completed bookings. Trade-off: no booking history, and customers need their booking ID to manage a booking.
- **Hold a slot for 5 minutes while the customer books.** This stops two people booking the same slot. Trade-off: a slot can look taken for up to 5 minutes if someone leaves.

## How we would build it

I would build this as an MVP, the smallest version that removes the manual work, and improve it with real booking data. The team works in Scrum-style Agile, in sprints of two weeks, and each sprint ends with a demo that the owner can try. Figure 2 shows the stages.

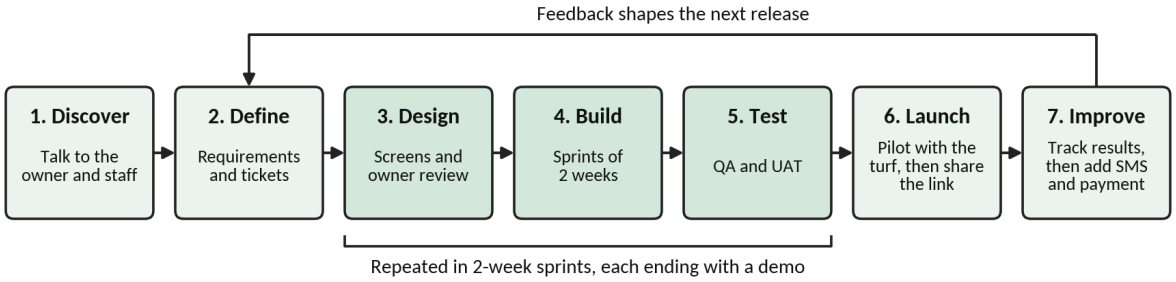



*Figure 2. Delivery approach (Scrum-style Agile)*

1. **Discover:** talk to the owner and staff, see how bookings work today and get answers to the questions above.
2. **Define:** write requirements and Jira tickets with acceptance criteria.
3. **Design:** the designer prepares screens for a phone, and the owner reviews them.
4. **Build:** engineers build in sprints, starting with availability and booking, then cancellation, then the admin page.
5. **Test:** QA runs the test cases, and the owner and a few real customers do user acceptance testing (UAT).
6. **Launch:** run alongside phone and WhatsApp bookings for two weeks, then share the link on the turf's Facebook page.
7. **Improve:** track the results below, then add SMS and online payment.

### Working with design and engineering

- **Design:** share the user flow and priorities first, review the screens against the acceptance criteria, and keep them simple, because most customers will use a phone.
- **Engineering:** go through each ticket before the sprint, discuss edge cases (double booking, hold expiry, cancellation time), agree what fits in the sprint and decide trade-offs together. Choices such as framework and hosting belong to the engineers, and I check them against the budget and timeline.
- **QA:** test cases come from the acceptance criteria, so everyone shares the same meaning of done.

## How I would measure success

Targets would be agreed with the owner after checking today's numbers.

- **Online share:** the percentage of bookings made through the system. The goal is that most bookings come from it within two months.
- **Double bookings:** zero.
- **Time to book:** under one minute from opening the page to confirmation.
- **Staff effort:** fewer booking calls and messages for staff each week.
- **Slot use and no-shows:** the percentage of slots booked, especially off-peak, plus no-shows and late cancellations, to decide when advance payment is needed.

## User flow

The main customer path is: View Turf, Select Date, View Available Slots, Select Slot, Enter Details, Confirm Booking, Booking Confirmed. Figure 3 shows the full flow, including cancellation.

### The selected slot is already booked

Booked slots are shown as disabled and cannot be selected. If someone else confirms the same slot while the customer is still typing, the system checks the slot again at confirmation, shows a message that the slot was just booked and lists the nearest free slots. The customer's details are kept.

### The customer cancels

Before confirming, the held slot is released after 5 minutes, or straight away if the customer goes back. After confirming, the customer opens Manage Booking with the booking ID and mobile number. If the slot starts in more than 12 hours, the booking is cancelled, the slot becomes free for other customers and a confirmation is shown. Otherwise the system says the booking cannot be cancelled online and shows the turf's phone number.

### Something goes wrong during booking

If details are missing or invalid, a message appears and the entered details are kept. If there is a system or network error, the customer sees that the booking was not confirmed and can retry. A retry never creates a duplicate booking. If the turf is closed or under maintenance, the date or slot shows as unavailable with the reason. If the owner blocks a slot that already has a booking, the admin sees a warning and contacts the customer.

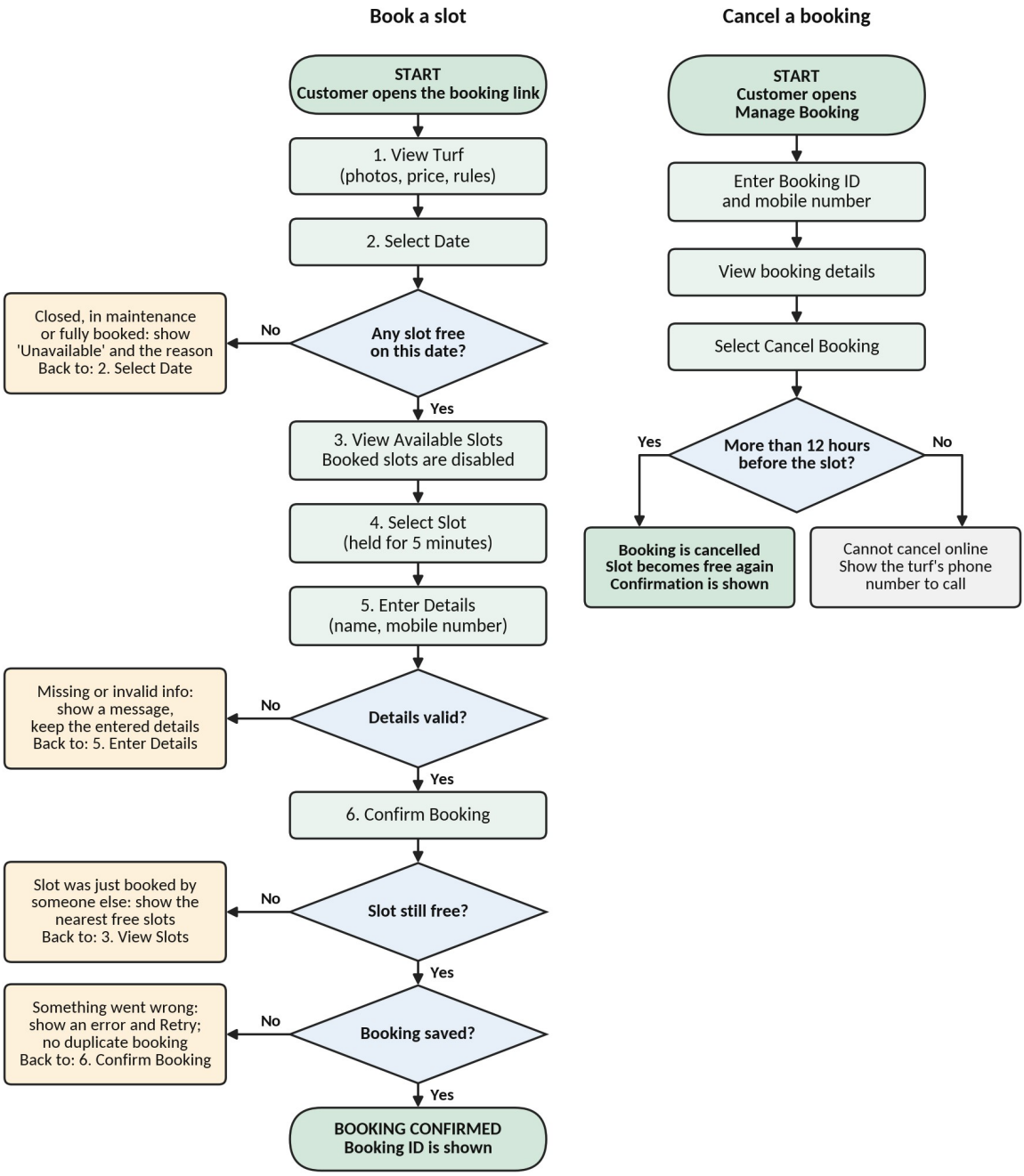



*Figure 3. Customer user flow*